

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (IV)	Branch: Computer Science & Engineering
Subject: Object Oriented Programming (with JAVA)	Course Code: B022414(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 2 T: 1 P: 0 Credits: 3
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

UNIT- I Introduction & Fundamentals of JAVA, Background of JAVA, About Java Technology, Java's architecture, Reading console inputs, Arrays, Constructors, Finalize method, final, this method and reference, static members.

UNIT-II Concrete class, Abstract class, Interface, Inner classes. Aggregation, Composition and Inheritance, super method and reference. Method overloading and overriding. Singleton classes. Package concepts. Exception Handling: Inbuilt, User defined, Checked and Unchecked.

UNIT- III String class. Wrapper classes (Integer, Boolean, Character, etc.). Multi-threading: Thread concept, Thread class, Runnable interface, Creating customized threads, Thread synchronization, Thread class methods. Java I/O: Use of InputStream, OutputStream, Reader and Writer classes for reading from and writing data into disk files.

UNIT-IV AWT & SWING: Frame, Panel, Dialog, CheckBox, Choice, List, JComboBox, JFrame, JPanel, JRadioButton, JScrollPane, JTabbedPane, Using Listeners: ActionListener, ContainerListener, FocusListener, ItemListener, KeyListener, MouseListener, TextListener, WindowListener. Applets. JDBC: Type1 to Type4 drivers. Java Networking: ServerSocket, Socket, RMI.

UNIT-V Collections Frameworks: HashSet, TreeSet, ArrayList, LinkedList, Vector, HashMap, TreeMap, Hashtable classes. Generics in Java: Creating instances of generic classes, generic types, Declaring (and invoking) methods that take generic types. Creating and running executable JAR (Java ARchives).

Text Books:

1. Herbert Schildt: "Java A Beginner's Guide, 7th edition", Oracle Press.
2. Maurice Naftalin, Philip Wadler, "Javas Generics and Collections", O'Reilly Media, Inc.
3. Benjamin J Evans, David Flanagan, "Java in a Nutshell", O'Reilly Media, Inc.

Reference Books:

1. Kathy Sierra, Bert Bates, "Head First Java", O'Reilly Media, Inc.
2. Kathy Sierra, Bert Bates, "OCA Java SE 8 Programmer I Exam Guide", McGraw Hill Professional.
3. Kathy Sierra, Bert Bates, "OCA Java SE 8 Programmer II Exam Guide", McGraw Hill Professional.
4. Kathy Sierra, Bert Bates, "OCA/OCJP Java SE 8 Programmer Certification Bundle", McGraw Hill Professional.

Course Outcomes [The students should be able to]:

1. Apply Java in developing Object Oriented code.
2. Apply the knowledge of Multi-threading and Streams in developing Java applications.
3. Design and implement applications using GUI and Networking in Java.
4. Apply the knowledge of Collections and Generics for building Java applications.
5. Design and develop Java based applications for solutions to real world problems.

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Roll No. :



B022414(022)

**B. Tech. (Fourth Semester) Examination,
April-May 2025**

(Scheme : AICTE)

(CSE Branch)

OBJECT ORIENTED PROGRAMMING (with Java)

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) of each question is compulsory carrying 4 marks. Attempt any two parts from part (b), (c) and (d) carrying 8 marks each.

Unit-I

1. (a) Describe java architecture.

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- (b) Define java token. Explain various statements defined in java with example.
- (c) Explain Local, Instance and static variable with suitable example.
- (d) How to take reading from console inputs. Explain with example.

Unit-II

- 2. (a) Describe Concrete class.
- (b) Explain checked and unchecked exceptions in Java and demonstrate them, using a Java program.
- (c) Explain the use try, catch, throws and finally with proper example.
- (d) Differentiate interface and abstract class.

Unit-III

- 3. (a) What are the different types of wrapper classes in Java?

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- (b) Explain thread synchronization with example.
- (c) Show the use of InputStream and OutputStream classes.
- (d) Differentiate between string and string builder. Briefly explain string class methods.

Unit-IV

- 4. (a) What is RMI?
- (b) What is JDBC? Explain the types of Drivers in JDBC.
- (c) Explain Event Delegation Model in Java with suitable example.
- (d) Explain layout manager. Also give example of any one layout.

Unit-V

- 5. (a) Describe Array list with example.

- (b) What is Linkedlist? give difference between Arraylist and Linkedlist?
- (c) What are the steps to creating and running executable JAR?
- (d) Explain Generics class in Java.

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Roll No. :

B022414(022)

**B. Tech. (Fourth Semester) Examination,
April-May 2022**

(Scheme : AICTE)

(CSE Branch)

OBJECT ORIENTED PROGRAMMING (with Java)

Paper : (BT3020)

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) of each question is compulsory carrying 4 marks. Attempt any two parts from part (b), (c) and (d) carrying 8 marks each.

Unit-I

1. (a) Compare Java and C++.
- (b) Explain static and this keyword with suitable example.

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- (c) Explain the features of Java.
- (d) Define constructor. Explain types of constructor using example.

Unit-II

- 2. (a) Differentiate between interface and abstract class.
- (b) Write a Java Program to demonstrate how packages can be created, imported and used.
- (c) Create a user defined exception class which throws exception when the user input marks greater than 100.
- (d) Differentiate between method overloading and method overriding with example.

Unit-III

- 3. (a) Compare string and stringbuilder.
- (b) Explain the life cycle of thread. In How many ways can a thread be created in Java? Explain with example.
- (c) Define synchronization. Explain the use of synchronization method with example.

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- (d) Explain Byte streams and Character streams with suitable Java program.

Unit-IV

- 4. (a) Write the differences between Swing and AWT components.
- (b) Write a Java Program to see the port number, protocol name of given URL.
- (c) Explain Event delegation model in Java with suitable example.
- (d) What is JDBC? Explain the types of JDBC.

Unit-V

- 5. (a) Define HashSet and TreeSet.
- (b) Explain generic using classes and generic types in Java.
- (c) Write a program using the concept of HashMap.
- (d) Explain Hashtable classes in java with example.

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Roll No. :

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B. Tech. (Fourth Semester) Examination

April-May 2021

OBJECT PROGRAMMING (with Java)

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Questions no. (a) is compulsory and attempt any two questions from (b), (c) and (d).

Unit-I

1. (a) Give the name of top class of all the classes in java
and top package which is default package. 4
- (b) Java is not a 100% pure object oriented programming
language. Explain with suitable example. 8

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[2]

- (c) Write a program taking input from user 1 to 3 and do the following : 8
- (i) If the input is 1 call the default constructor.
 - (ii) If the input is 2 call the constructor with one integer parameter.
 - (iii) If the input is 3 call the constructor with one string parameter.
- (d) What is JVM? Also explain the role of Garbage Collector with suitable program. 8

Unit-II

2. (a) Give uses of super in Java. 4
- (b) Differentiate between Abstract class and Interface with example. 8
- (c) Find the output of the two following code : 8
- (i) public class test

```
{
    public static void main (String [ ] args)
    {
        try
        {
```

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```
System.out.printf("1");
int sum = 9 / 0;
System.out.printf("2");
}
catch (Arithmetic Exception e)
{
    System.out.printf("3");
}
catch (Exception e)
{
    System.out.printf("4");
}
finally
{
    System.out.printf("5");
}
}
(ii) public class test
{
    private void m1()
    {
        m2 ();
```

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[4]

```

        System.out.printf("1");
    }
    private void m2 ()
    {
        m3 ();
        System.out.printf("2");
    }
    private void m3 ()
    {
        System.out.printf("3");
        try
        {
            int sum = 4 / 0 ;
            System.out.printf("4");
        }
        catch (Arithmetic Exception e)
        {
            System.out.printf("5");
        }
        System.out.printf("7");
    }
    public static void main (String [ ] args)
    {

```

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[5]

```

Test obj = new Test ();
obj.m1 ();
}

```

(d) Write a program in Java to demonstrate user throw.

Also explain checked throw. Also explain checked and unchecked exception. 8

Unit-III

3. (a) Which one is the master class of exceptions? 4
- (b) Explain the concept of Lock in java. Also explain Thread Priority. 8
- (c) Explain string and string buffer class. What is mutable and immutable in java? 8
- (d) Explain Inter Thread Communication. :What is Daemon Thread? 8

Unit-IV

4. (a) What is the functioning of who is? 4
- (b) Write a program to add 2 numbers using Applet. 8

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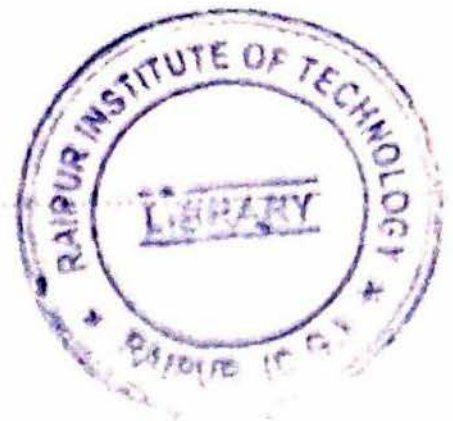
PTO

[6]

- (c) Explain all types of JDBC Bridge drivers. 8
- (d) Write a program using inter address class and show the utility of URL and URL connection classes. 8

Unit-V

5. (a) What is AWT? 4
- (b) Explain Event Delegation Model. 8
- (c) How do you retrieve all keys present in a Hash Map? 8
- (d) Write the steps of creating and running executable JAR files in java. 8



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B. Tech. (Fourth Semester) Examination,

Nov.-Dec. 2024

(New Scheme)

(CSE Branch)

OBJECT ORIENTED PROGRAMMING (with Java)

Paper : (BT3020)

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) is of 4 marks and part (b), (c) and (d) are 8 marks each. Part (a) is compulsory and attempt any two parts from (b), (c) and (d). Assume any missing data.

Unit-I

1. (a) Describe the use of static member in JAVA.

4

[2]

- (b) Define and describe the rules of constructor and also explain copy constructor. 8
- (c) Why we use this method and reference? Explain with suitable JAVA program. 8
- (d) Explain the following : 8
- (a) JAVA Architecture
- (b) JAVA fundamental and background

Unit-II

2. (a) Compare concrete and abstract class. 4
- (b) Explain the concept of exception handling with suitable JAVA program. 8
- (c) Write JAVA program to explain super method and reference. 8
- (d) Write JAVA program for : 8
- (i) Inner Class
- (ii) Method overloading

Unit-III

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[3]

3. (a) Compare the use of Input Stream and Output Stream. 4
- (b) Describe Thread synchronization and Thread class methods using suitable example. 8
- (c) Write a JAVA program to explain the concept of reading and writing data from/to disk. 8
- (d) Define and explain Wrapper class and string class with JAVA program. 8

Unit-IV

4. (a) Define and describe Applet. 4
- (b) Write a JAVA program to explain Server Socket. 8
- (c) Explain different AWT and SWING tools with suitable example. 8
- (d) Explain the following. 8
- (i) Type 1 to Type 4 drivers
- (ii) Action Listener and Container Listener

Unit-V

5. (a) Write the use of Hashset and Hashtable class. 4

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- (b) Write JAVA program to create and execute JAVA Archives. 8
- (c) Explain the collection framework with suitable examples. 8
- (d) Explain with code the concept of Generic classes & methods. 8



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**B. Tech. (Fourth Semester) Examination,
April-May 2024
(New Scheme)**

**(Computer Science Engineering Branch)
OBJECT ORIENTED PROGRAMMING
WITH JAVA**

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions, Part (a) from each question is compulsory and carry 4 marks. Attempt any two parts from parts (b), (c) and (d) of each question carrying 8 marks.

1. (a) How Java is platform independent?
- (b) How do you set up the environment in java?
- (c) Java is not 100% object oriented language.

Comment with suitable example.

- (d) Write a program to implement constructor in java?
- 2. (a) What is singleton class in java?
- (b) Demonstrate arithmetic exception in java?
- (c) What is the difference between method overloading and overriding?
- (d) What is abstract class? Write a program to implement abstract class in java?
- 3. (a) What is a wrapper class?
- (b) Write a program to create a thread using runnable interface.
- (c) Write a program to demonstrate reading and writing to files in java.
- (d) Explain inputstream and outputstream in detail with suitable example.
- 4. (a) Explain applet tag with all its attribute in detail.
- (b) Write a program in java to create user form?
- (c) Define sockets. Write a program to demonstrate

simple TCP client-server interaction.

- (d) Explain the following in nutshell :
 - i. ActionListener
 - ii. ContainerListener
 - iii. FocusListener
 - iv. ItemListener
- 5. (a) What are generic classes? What is the purpose of generic classes in java?
- (b) Differentiate between HashMap and Hashtable.
- (c) What is JAR file? How to create and execute JAR file in Java.
- (d) Explain the following in nutshell :
 - i. HashSet
 - ii. TreeSet
 - iii. ArrayList
 - iv. LinkedList



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**B. Tech. (Fourth Semester) Examination,
April-May 2023**

(Scheme : AICTE)

(CSE Branch)

OBJECT ORIENTED PROGRAMMING (with Java)

Paper : (BT3020)

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) of each question is compulsory carrying 4 marks. Attempt any two parts from part (b), (c) and (d) carrying 8 marks each.

Unit-I

1. (a) Give any 4 differences between C++ and Java.

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- (b) Explain Constructors in Java with the help of example.
- (c) Explain static members in java with proper example.
- (d) What is Java Machine? Explain its architecture.

Unit-II

2. (a) Write down any 4 differences between method overloading and overriding.
- (b) Explain Exception-Handling in Java. Give one example for user defined exception.
- (c) Explain Abstract Class and Interface with suitable example.
- (d) Discuss about Singleton Class in Java using example.

Unit-III

3. (a) What is multithreading?
- (b) What is synchronization in Java? Explain using suitable examples.

- (c) Explain following :
 - (i) Reader Class
 - (ii) Writer Class
- (d) What is mutable and immutable in Java. Also explain string and string buffer class.

Unit-IV

4. (a) Explain AWT in Java.
- (b) Explain the life cycle of Applets using example.
- (c) Explain JDBC drivers available in Java.
- (d) Explain Server Socket with the help of example.

Unit-V

5. (a) What do you understand by Java Collection Framework?
- (b) Explain following :
 - (i) Array List
 - (ii) Linked List

(c) How do you get all the keys present in a Hash Map?

(d) Explain Event Delegation Model (EDM).



B022412(022)

B. Tech (Fourth Semester) Examination,

April-May 2023

(AICTE Scheme)

(CSE Branch)

COMPUTER SYSTEM ARCHITECTURE

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions. Part (a) in each unit is compulsory. Attempt any two from (b), (c) and (d). Part (a) is of 4 marks and part (b), (c) and (d) are of 8 marks each.

Unit-I

1. (a) What is difference between :

4

(i) MAR and MDR

(ii) General purpose register and instruction register

(b) Consider the following possibilities for saving a return for subroutines :

- (i) In a processor register
- (ii) In a memory location associated with the call, so that a different location is used when the sub-routine is called from different places.

(iii) On a stack

Which of these possibilities supports subroutine nesting and which supports subroutine recursion? Explain each. 8

(c) Analyze various buses used in bus structure and express the importance of all buses. 8

(d) What is Addressing Mode? Explain any 3 addressing modes. An instruction is stored at location 300 with its address field at location 301. The address field has the value 400, a processor register R1 contain the number 200. Evaluate the effective address if the addressing modes of instruction are : 8

- (i) Direct,
- (ii) Immediate,

- (iii) Relative,
- (iv) Register indirect

Unit-II

2. (a) What is a Fast Adder? Design a 4 bit adder with carry-lookahead. 4
- (b) Define bit pair recoding. Explain Booth's multiplication alongwith for 2's complement number using flowchart and example. 8
- (c) Design datapath for an n -bit two's complement adder subtractor. 8
- (d) Define guard bits. Describe the implementation of binary parallel adder/subtractor circuit using full adders. 8

Unit-III

3. (a) What is Virtual Memory? Write its advantages and disadvantages. How is it different from main memory? 4
- (b) Draw and explain Memory Hierarchy. Explain each level in brief. 8

(c) A block set-associative cache memory consists of 12 blocks divided into 4 block set. The main memory consists of 16,384 blocks and each block contains 256 eight bit words.

(i) How many bits are required for addressing the main memory?

(ii) How many bits are needed to represent the TAG, SET and WORD fields? 8

(d) Explain cache mapping. Explain the types of cache mapping with example. 8

Unit-IV

4. (a) Differentiate between synchronous and asynchronous data transfer. 4

(b) What is Address space? Explain Isolated I/O and Memory Mapped I/O. 8

(c) What is DMA transfer? Explain role of DMA controller with diagram. 8

(d) Write short notes on :

(i) Interrupt handling mechanism

(ii) Fault tolerance systems 8

Unit-V

5. (a) Consider a pipeline having 4 phase with duration 60, 50, 90 and 880 ns. Given latch delay is 10 ns. Calculate :

(i) Non-pipeline execution time

(ii) Speed-up ratio 4

(b) What is Parallel Processing? Explain Flynn's classifications of parallel architecture. 8

(c) What is Pipelining? Explain 4 segment of instruction pipeline. 8

(d) What is Arithmetic Pipeline? Explain floating point addition using arithmetic pipeline. Also write its applications. 8